QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	*RESA — AI Business Strategist for Malaysian Online SMEs (UMHackathon 2026 — Team 4Flat)*
Team Repo URL	*https://github.com/ejafee/4flat*
Project Board URL	*https://trello.com/invite/b/69eccb11176c9594e2daf8cb/ATTI0878f3f526c4b718dffbb06bd67e223cD7154534/4flat-um-hackathon-26*
Live Deployment URL (optional for preliminary round)	*To be deployed before final round submission*

Objective:

The primary objective of this QATD is to ensure that RESA (Real-time Economic Strategy Advisor) reliably accepts business inventory data and unstructured context from the user, processes it through the GLM integration (via ilmu.ai), and returns a coherent, specific, and actionable marketing and business strategy. Testing must verify that RESA behaves correctly across normal, edge, and adversarial inputs, and that the AI output meets a defined quality threshold for strategy completeness and accuracy.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

*This section aligns the testing connected back to specific user requirements via **Requirement Traceability Matrix".** So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.*

1.1 In-Scope Core Features

- Inventory Input: User manually enters product name, cost price, sell price, stock quantity, and units sold via dynamic form rows.
- Business Context Input: Free-form textarea where the owner describes their current situation (e.g. upcoming sale events, cash pressure, competitor actions)
- File Upload (Optional): User uploads a CSV or Excel export from Shopee, Lazada, or TikTok Shop for richer data context .
- Internal Dataset Loading: RESA automatically reads all CSV/Excel files from the datasets/ folder and injects them as market intelligence into the AI prompt.

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- AI Strategy Generation: GLM processes all structured and unstructured inputs and returns a formatted 7-section strategy report.
- Result Display: The strategy output is rendered in result.html with section headers styled in amber/orange, a copy-to-clipboard button, and an inventory recap table

1.2 Out-of-Scope

- User authentication and account management
- Persistent storage of past strategies or user history
- Real-time Shopee/Lazada API integration (users export manually)
- Mobile-native application (web only for this submission).

2. Risk Assessment & Mitigation Strategy

Quality Assurance risks are anticipatory. Technical risks have been identified based on RESA's architecture (Flask + Pandas + GLM API via proxy) and evaluated using the 5×5 Risk Assessment Matrix. Risk Score = Likelihood × Severity.

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
GLM API timeout / incomplete response (proxy drops connection mid-stream)	4	4	16 (Critical)	Switched from streaming to blocking request with 180s read timeout via custom httpx client. Fallback: Gemini API as secondary provider.	Manually trigger slow/large prompts and measure response completion. Verified fix resolves 'incomplete chunked read' error
Uploaded file with unexpected columns breaks data_handler parsing	3	3	9 (Medium)	parse_inventory_file() uses try/except on all pandas operations. Invalid files return graceful error message, not a 500 crash.	Upload malformed CSV, Excel with missing headers, and wrong file types (.txt, .pdf). Confirm error message shown and app remains stable.

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Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
GLM hallucinates or returns incomplete strategy (missing sections)	3	4	12 (High)	System prompt enforces strict 7-section output format with exact emoji headers. Temperature set to 0.7-0.8 to balance creativity and reliability.	Run AI-01, AI-02, AI-03 prompt pairs. Verify all 7 section headers present in output. Flag if any section is missing or contains placeholder text.
Large internal datasets slow down response time significantly	2	3	6 (Medium)	load_internal_datasets() reads only df.head(5) per file, not full dataset. Silently skips unreadable files.	Add 3 large CSV files (10,000+ rows each) to datasets/ folder. Measure total response time. Target: dataset loading adds <2s to total time.
User submits form with no inventory items (empty table)	3	2	6 (Medium)	app.py validates that at least 1 inventory item exists before calling GLM. Returns user-friendly validation error if empty.	Submit form with all inventory rows removed. Confirm error message shown and no API call is made (check server logs).

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

Risk Level Reference:

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Risk Score	***Risk Level***	***Recommended Action***
1 – 5	*Low*	*Monitor only. Acceptable risks.*
6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

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3. Test Environment & Execution Strategy

This section explains where and how RESA is tested, how test data is managed, and the pass/fail thresholds governing each testing phase.

- **Unit Test**
    - Scope: Individual Python functions — parse_inventory_file(), build_inventory_summary(), build_user_prompt(), get_strategy() error handling.
    - Execution: Manual unit tests written in pytest. Run locally during development on each function change
    - Isolation: External API calls (GLM client) are mocked using unittest.mock. Pandas file parsing tested with synthetic CSV/Excel fixtures.
    - Pass Condition: All happy path inputs return correct data types and structure. Negative inputs (empty file, wrong extension) return expected error strings without raising uncaught exceptions
- **Integration Test**
    - Scope: Full flow from form submission (/analyse route) through data_handler, prompt_engine, and glm_client to result rendering.
    - Execution: Manual end-to-end test performed after each significant code change. Test with real API call to confirm full pipeline connectivity.
    - Workflow: Real API calls made (no mocking at integration level) to validate actual GLM response is received and rendered correctly in result.html.
    - Pass Condition: result.html renders with all 7 strategy sections present, inventory recap table shows correct margin calculations, and copy button functions.
- **Test Environment (CI/CD Practice):**
    - Local Testing: All development testing performed manually on localhost:5000. Developer runs flask run and tests via browser.
    - Version Control: GitHub repository used for all code management. Feature branches created per component, merged to main via pull request.
    - CI/CD Plan (Intended): GitHub Actions workflow planned to trigger pytest unit tests on every push to any branch. PR to main branch requires all unit tests to pass before merge is allowed. Not yet fully configured at preliminary round stage.

- **Regression Testing & Pass/Fail Rules:**
    - Execution Phase: Regression test run manually across all 5 in-scope features each time a new component is merged to main.
    - Pass/Fail Condition: A test case passes only when the actual output matches the expected output as defined in Section 5. Any mismatch is logged as a defect.
    - Continuation Rule: Critical tests (TC-01 Happy Case) must pass before NFR tests (TC-03, TC-04) are executed. A broken happy path invalidates performance metrics.

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- **Test Data Strategy:**
    - Manual: Three business scenario datasets prepared — F&B (Kak Aina Cookies), Fashion (ZoneX Streetwear), and a blank/adversarial input (???) — used as standard test fixtures.
    - Internal Datasets: 4 CSV files in datasets/ folder (Fashion_Retail_Sales.csv, Womens Clothing E-Commerce Reviews.csv, TikTok scraper data, Online Sales Data.csv) serve as background market intelligence.
    - Uploaded File Test: A sample Shopee order export CSV used to test the optional file upload flow.
    - A minimum of 85% of all executed test cases must pass for preliminary round submission.
    - All Critical test cases (TC-01 Happy Case, AI-01) must pass at 100%. No exceptions.
    - AI Output tests: At least 2 of 3 prompt/response pairs (AI-01, AI-02, AI-03) must meet their defined acceptance criteria.

- **Passing Rate Threshold:**
    - A minimum of 85% of all executed test cases must pass for preliminary round submission
    - All Critical test cases (TC-01 Happy Case, AI-01) must pass at 100%. No exceptions
    - AI Output tests: At least 2 of 3 prompt/response pairs (AI-01, AI-02, AI-03) must meet their defined acceptance criteria.

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The main branch must remain a stable source of truth. These checks must pass before any branch is merged.

Checks	Requirements	Project	PASS/FAIL
Automatic Build	Zero build errors when running flask run	0 Errors	**PASS**
Unit Tests	100% of written unit tests passing	100% (4/4 functions tested)	**PASS**

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Code Quality	Zero critical linting errors (flake8)	Minor style warnings only (E501 line length)	**PASS**
Test Coverage	Minimum 70% coverage on core modules	~72% (data_handler, prompt_engine covered)	**PASS**

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4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements	Project	PASS/FAIL
Automatic Build	Minimum 85% of integration tests pass	87.5% (7/8 test cases passed)	**PASS**
AI Output Pass Rate	Minimum 66% of documented prompt/response pairs meet acceptance criteria	100% (3/3 AI test pairs passed)	**PASS**
Critical Bugs	Zero P0 bugs (app-breaking issues)	1 resolved (API timeout — fixed via httpx timeout config)	**PASS**
API Response Time	Strategy generated within 60s under normal conditions	Avg 30s	**PASS**
Security	Security	Clean — .env in .gitignore confirmed	**PASS**

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5. Test Case Specifications (Drafts)

Test cases cover the 4 required categories: 1 Happy Case (entire flow), 1 Negative/Edge case, and 2 Non-Functional (performance/load) tests.

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	**Happy Case (Entire Flow):** Full strategy generation from form submission to result page	Verify that a user can fill in business info, add inventory items, write a situation, submit the form, and receive a complete 7-section AI strategy on the result page	1. Open RESA at localhost:5000 2. Enter Business Name: Kak Aina Cookies 3. Enter Business Type: F&B 4. Write situation: 'Raya in 2 weeks, 450 units unsold' 5. Add 3 inventory items with cost/sell/stock/sold 6. Click Generate Strategy 7. Wait for result.html to load	Result page loads with all 7 sections present. Strategy references actual inventory items and RM figures. Inventory recap table shows correct margin calculations. Copy button functional..	Strategy generated successfully in 24s. All 7 sections present. RESA recommended a 'Set Lengkap Raya' bundle at RM18 with exact RM projections. Inventory table rendered correctly. **Status: PASS**

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-02	**Negative Case:** Invalid file upload (wrong file type)	Verify that uploading a non-CSV/Excel file (e.g. .txt or .pdf) is rejected gracefully without crashing the app or calling the AI API	1. Open RESA at localhost:5000 2. Fill in business name and type 3. Add at least 1 inventory item 4. In the file upload zone, upload a .txt or .pdf file 5. Click Generate Strategy	App redirects back to index with a flash error message: 'Invalid file type. Please upload a CSV or Excel file.' No API call is made. App remains stable with no 500 error	App redirects back to index with a flash error message: 'Invalid file type. Please upload a CSV or Excel file.' No API call is made. App remains stable with no 500 error. **Status: PASS**
TC-03	**NFR (Performance):** API response time under normal single-user load	Verify that RESA returns a complete strategy within 60 seconds under a standard single-user request with a typical inventory size (3-5 items).	1. Open RESA with browser dev tools Network tab open 2. Submit a standard form (3 items, business context filled) 3. Record time from form submission to result.html fully loaded 4. Repeat 5 times and record average	Average response time < 60 seconds. 0% error rate across 5 runs. Result page loads completely with full strategy content.	Average response time < 60 seconds. 0% error rate across 5 runs. Result page loads completely with full strategy content. **Status: PASS**

6. AI Output & Boundary Testing (Drafts)

This section validates that RESA's GLM integration produces acceptable output quality and handles abnormal inputs gracefully

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6.1. **Prompt/Response Test Pairs**

Document at least 3 Prompts/response test pairs. For each of them, do define the acceptance criteria - what a correct, acceptable or a failing response may look like for your use case.

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	This section validates that RESA's GLM integration produces acceptable output quality and handles abnormal inputs gracefully.	Output must: (1) identify the correct cash risk, (2) recommend a specific bundle with RM pricing, (3) generate at least 1 ready-to-post social media caption in Manglish, (4) include a 7-day content plan, (5) contain no hallucinated product names.	RESA identified RM1,525 locked in unsold stock, Recommended 'Set Lengkap Raya' bundle at RM18 (cost RM9.50, 44,7% margin). Generated full 7-day TikTok/IG content rotation with Manglish captions. Revenue projection table: Conservative RM1,770 to Optimistic RM2,814. All 7 sections present. No hallucinations detected.	**PASS**
AI-02	ZoneX Streetwear. 6.6 sale in 5 days. 80 Black Hoodies barely selling (12 sold). Competitor dropped prices 20%. 3 inventory items with different margins.	Output must: (1) correctly compute RM2,800 in locked hoodie stock, (2) recommend a specific 6.6-themed pricing strategy, (3) not simply match competitor discount, (4) include a day-by-day 6.6 campaign plan, (5) all RM figures	RESA identified RM2,800 locked in hoodies. Recommended RM66 flash deal (30 units, 47% margin) + ZoneX Starter Pack bundle at RM99 (49.5% margin). Explicitly reasoned against matching competitor price. 7-day countdown	**PASS**

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		must be mathematically correct.	campaign generated with TikTok LIVE strategy. Revenue projection: RM5,294 total for 6.6 week. All figures verified correct.	
AI-03	Adversarial: Business name '???', type 'idk', situation 'help'. Single inventory item: name 'thing', all values RM0, 0 stock, 0 sold.	Output must: (1) not crash or return an error, (2) not hallucinate fake RM figures or products, (3) acknowledge lack of data honestly, (4) provide genuinely useful guidance for a zero-data situation, (5) not generate a promo strategy based on empty data.	RESA correctly identified zero usable data. Did not hallucinate any RM figures or product names. Pivoted to a 'Startup Validation Sprint' strategy with a generic product selection framework. Provided pricing formula (2.5x cost), tracking spreadsheet template, and 7-day launch plan applicable to any business. Tone remained helpful, not robotic.	**PASS**

6.2. Oversized/Larger Input Test

Fields	Details
Maximum Input Size	~20 inventory items + 500-word business context + 1 uploaded file (max 5MB). Estimated ~6,000-8,000 tokens total input.
Input used while testing	15 inventory items + 400-word context + 3 internal CSVs (12MB total) + 1 uploaded Shopee export (250KB). Estimated ~7,200 tokens.
Expected Behavior	GLM accepts input and returns complete strategy. Response time may increase. No truncation of output.
Actual Behavior	Strategy generated successfully. Response time 44s (within 60s threshold). All 7 sections present. Internal datasets summarised correctly (only head(5) rows per file, keeping token count manageable).

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Status	**PASS**

6.3. Adversarial/Edge Prompt Test

Tested with AI-03 (see Section 6.1). Input was intentionally malformed: business name '???', type 'idk', situation 'help', and a single inventory item with all-zero values.

GLM Response Behaviour:

- RESA did not crash or return a 500 error.
- The AI correctly identified that no actionable data was present and did not fabricate RM figures.
- Instead of generating a fake promo strategy, RESA pivoted to a generic startup framework — which is the correct behaviour for a zero-data scenario.
- The response tone remained warm and helpful, consistent with the system prompt's 'calm expert friend' instruction.

System Handling:

- The Flask /analyse route accepted the submission (no server-side crash).
- prompt_engine correctly injected the zero-value inventory into the prompt without throwing a division-by-zero error in build_inventory_summary() (margin calculation handles zero sell price via conditional check).

6.4. Hallucinating Handling

RESA mitigates hallucination risk through the following mechanisms:

- Grounded Prompt Design: The system prompt explicitly instructs the GLM to reference the owner's actual numbers (prices, margins, stock, context) in all reasoning. Generic advice is explicitly prohibited by the rule: 'Never give vague advice like post more often.'
- Internal Dataset Anchoring: Real market data (Fashion_Retail_Sales.csv, TikTok trends, e-commerce reviews) is injected as context, giving the model factual anchors to reference instead of generating data from memory.
- Mathematical Verification: All RM figures in AI-01 and AI-02 outputs were manually verified post-generation. Bundle pricing, margin percentages, and revenue projections were cross-checked against input data. No arithmetic errors found.
- Output Structure Enforcement: The 7-section format with exact emoji headers acts as a structural constraint that reduces the model's tendency to wander into unsupported claims.

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- Temperature Setting: Temperature set to 0.7-0.8 — creative enough for natural language captions, constrained enough to keep financial reasoning consistent.

Known Limitation: RESA cannot fully prevent hallucination in trend intelligence sections, as the model may reference trends from training data rather than live data. This is a known trade-off accepted at this stage of development. Future mitigation: integrate a web search tool (e.g. Gemini's grounding feature) to anchor trend claims to real-time sources.

Important Note:

Following Agile principles, all testing in this document takes place iteratively throughout the development cycle, not solely at the end. Each feature was tested as it was built, with defects resolved before the next component was developed.

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